Post Lab Questions – Signal Generator and Oscilloscope

1. Define Ohm's Law

Ohm's Law states that the current flowing through a conductor is directly proportional to the voltage applied across it and inversely proportional to its resistance, provided the temperature and physical conditions remain constant. The mathematical expression is: $V = I \times R$, where V is voltage, I is current and R is resistance.

2. What is a Resistor?

A resistor is an electrical component that limits or controls the flow of electric current in a circuit. It provides resistance and helps protect circuit components from excessive current. The resistance value is measured in ohms (Ω).

3. How does a Resistor work?

A resistor works by opposing the flow of electric current. When voltage is applied across a resistor, it restricts the current according to Ohm's Law. As current flows through the resistor, some electrical energy is converted into heat energy, which reduces the current in the circuit.

4. Inference from Input and Output Waveforms on Oscilloscope

From the oscilloscope observation, the input and output signals are sinusoidal waves with the same frequency. However, the amplitude of the output voltage across each resistor is smaller than the input voltage due to voltage division in the series circuit. The waveform shape remains the same, which shows that resistors do not change the waveform shape but only reduce its amplitude.

5. Calculations for 7V Input Signal

a) Voltage Calculation (using oscilloscope volt/div):

If the oscilloscope shows 3.5 divisions vertically and the volt/div setting is 2 V/div:
Peak Voltage (V_p) = $3.5 \times 2 = 7$ V peak-to-peak (approx).

b) Frequency and Period Calculation:

If one complete wave occupies 5 horizontal divisions and time/div = 1 ms:
Time Period (T) = 5 × 1 ms = 5 ms
Frequency (f) = 1 / T = 1 / 0.005 = 200 Hz

c) Current Calculation using Ohm's Law:

For Experiment 2 total resistance:
R1 = 4.7 kΩ, R2 = 3.3 kΩ, R3 = 5.1 kΩ
RT = 4.7 + 3.3 + 5.1 = 13.1 kΩ
Peak Voltage Vp = 7/2 = 3.5 V
I = V / R = 3.5 / 13100 = 0.000267 A ≈ 0.267 mA